

On the right track

Prevention is better than cure. And it certainly makes sense to spend on averting train collisions than on compensation after tragedy strikes. Focussing the spotlight on Indranil Majumdar, inventor of the RACAS

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You might say that Indranil Majumdar, born into a family of physicians, has a natural instinct to save lives, and this has probably led to his inventing the Railway Collision Avoidance System (RACAS 2001), the first of its kind in the world.

Majumdar had his professional training in Electronics and Communications Engineering from the Institution of Engineers and has been working in this field for more than 16 years now. Going nostalgic about his childhood, he remembers that he was always inspired by inventors such as Einstein, Franklin, J.C. Bose and Edison. And in the same spirit, he is always trying

for his RACAS, he thought it was just someone pulling a prank on him.

RACAS incorporates a variety of high performance power management, interface, RF logic and microcontroller products. Existing collision systems deployed in developed nations such as Germany and Japan involve a routing system that is backed up by an efficient network management system.

RACAS, Majumdar says, is a unique system to avert collisions. The Indian Railways at present has only a traffic management system and no collision aversion system and RACAS can fill in this lacuna. He claims that RACAS, when programmed to suit the existing system of rail network, reduces the possibility of train collisions to nil.

The cost of implementing such anti-collision systems is far less than what the government spends on compensation in case of an accident. According to Majumdar, the full prototype and complete installation and testing would require at least 12 months. The inventor believes prevention is better than cure. He says that although we would all have reason to rejoice that we have a life saving invention from India, the real joy would be to see RACAS being put into effect to prevent any more railway disasters. "I am confident that given a chance to prove its worth, I can make RACAS talk", he says. Majumdar says that his collision aversion system is the fruit of his long-cherished desire to invent something that would serve society. Obviously, he would need the help of the Indian government in implementing this life saving system on the Indian rail network. Is anyone out there listening?

Majumdar, when not busy inventing, is an active member of the American Radio Relay League, IEEE (USA), IEEE Communications Society and IEEE Antennas and Propagation Society. He is also a ham and is a life member of the Amateur Radio Society of India.

"Not all people are ambitious, but some have a vision and they are out there to fulfil it," says Majumdar. He is neither ambitious nor does he seek monetary benefits, but he does believe in holding up the banner of technical excellence. His vision is to convert Fox Radio into a research and design centre and has plans to channel a part of the Texas Instruments award money of \$100,000 towards this effort. ■

"I am not looking for money, I crave technical excellence"

Indranil Majumdar, CEO, Fox Radio

something new.

Starting his career as an apprentice in the Birla Industrial and Technological Museum in 1985, Majumdar has had a rising career graph and today he is the CEO of Kolkata-based Fox Radio. He has a long list of awards which include the Motorola Discover an Inventor, 1998 prize and the Exceptional Service Award from the Disaster Preparedness and Emergency Response Association (DERA), the US, for his amateur radio disaster communications operation when Orissa was hit by a super-cyclone in 1999. All these years of experience have been a training ground of sorts for him.

However, when he received a call from Texas Instruments on July 29, 2001, informing him that he had won their Analog Design Challenge